

OPTICAL AND PERFUSION MEASUREMENT EXPANSION

Optical Perfusion Monitoring Concept

The wearable longitudinal penile physiology monitoring device may incorporate optical sensing modalities configured to detect physiologic characteristics associated with blood flow, vascular activity, and tissue perfusion. Optical measurements may provide contextual information complementary to mechanical rigidity sensing by evaluating vascular response patterns occurring during physiologic state transitions.

Optical sensing may operate by emitting light toward tissue and detecting reflected, scattered, or transmitted light signals whose characteristics vary according to blood volume, pulsatile flow, or tissue optical properties. Such measurements may contribute to estimation of perfusion dynamics, pulse characteristics, and vascular responsiveness.

Photoplethysmography (PPG) Embodiments

Photoplethysmography sensing may be implemented using one or more light emitters and one or more photodetectors positioned within the wearable structure. Emitters may include light-emitting diodes or other optical sources operating at single or multiple wavelengths. Multi-wavelength operation may allow differentiation of tissue optical responses and improved robustness against motion artifacts or environmental lighting conditions.

PPG sensing configurations may include reflective sensing arrangements, transmissive sensing arrangements, angled optical paths, distributed emitter arrays, or multi-detector geometries. Sampling may be continuous, periodic, burst-based, or triggered by detected physiologic events.

Optical Signal Stabilization Techniques

Optical sensing performance may be improved through mechanical isolation, conformal contact surfaces, optical shielding structures, adaptive signal filtering, motion-aware gating, or algorithmic noise reduction methods. Integration with inertial sensing may enable rejection of motion-corrupted data segments and enhancement of physiologic signal fidelity.

Perfusion and Vascular Response Modeling

Optical signals may be analyzed to derive pulse rate, pulse amplitude proxies, perfusion indices, vascular response timing, and temporal variability characteristics. Derived perfusion metrics may be combined with mechanical deformation measurements to improve interpretation of erection onset, stability, and recovery dynamics.

Perfusion modeling may incorporate time-domain analysis, frequency-domain analysis, statistical feature extraction, or machine learning interpretation frameworks capable of identifying patterns associated with physiologic activation and vascular regulation.

Alternative Optical and Bio-Optical Techniques

Alternative embodiments may employ optical spectroscopy, near-infrared sensing, laser-based sensing, speckle flow detection, optical coherence approaches, or other bio-optical measurement techniques capable of inferring vascular or tissue characteristics. Hybrid optical architectures combining multiple optical modalities are contemplated within the scope of this disclosure.

The optical sensing embodiments described herein are illustrative and non-limiting and encompass present and future optical and bio-optical technologies capable of contributing to longitudinal physiologic monitoring through wearable perfusion analysis.